Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	31 January 2025
Team ID	
Project Name	College Library Book Request System
Maximum Marks	4 Marks

Technical Architecture:

The College Library Book Request System uses a 3-tier web architecture. Students access a custom book request form through the web portal, the backend processes date validation and auto-generates a unique request number, and the librarian is notified automatically upon each submission.

System: College Library Book Request System — Students request books from the college library through a digital form with automated validation and librarian notification.

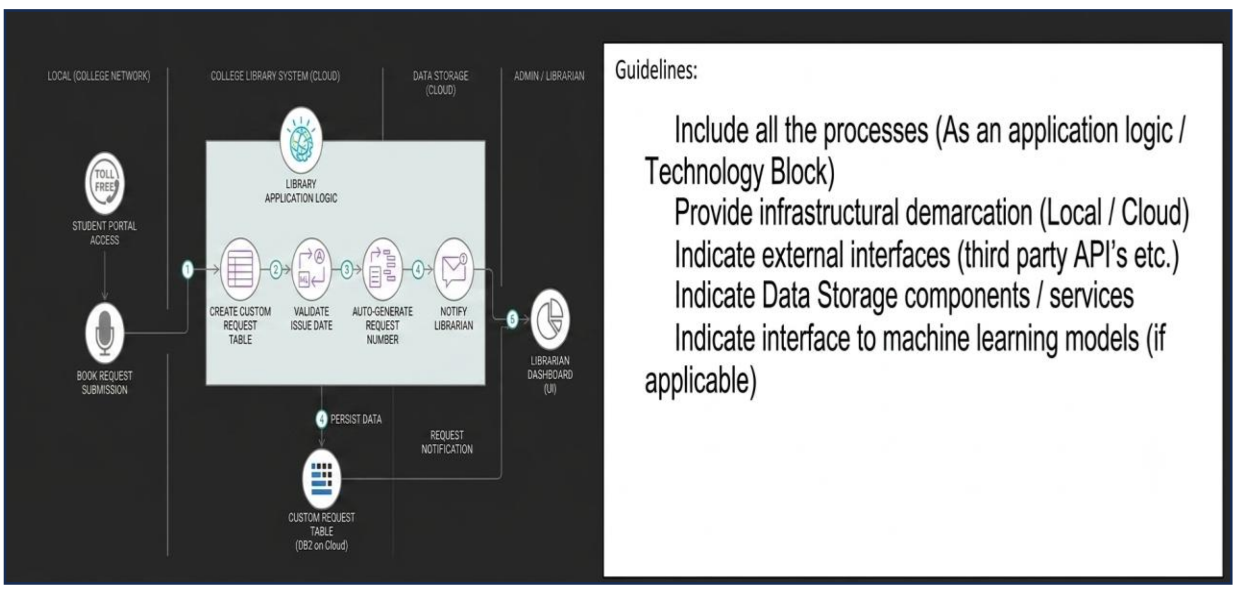

Table-1: Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web portal through which students submit book requests using a custom form	HTML, CSS, JavaScript / React JS

2	Book Request Form	Custom form to capture book title, author, student ID, and issue date with date validation	React JS / Angular JS with Form Validation
3	Application Logic — Request Processing	Backend logic to validate issue date, auto-generate request number, and store the request	Python (Flask / Django) or Node.js
4	Application Logic — Notification Service	Sends real-time notification to librarian when a new book request is submitted	Email API (SMTP / SendGrid) / WebSocket
5	Database	Stores book request records, student details, and request status	MySQL / PostgreSQL
6	Request Number Generator	Auto-generates a unique sequential request number (e.g. REQ-2025-0001) upon form submission	UUID / Sequential ID library
7	Authentication Module	Manages student and librarian login and access control	JWT / OAuth 2.0
8	Infrastructure (Server / Cloud)	Hosts the library request web application and database	Cloud (AWS / Azure) or Local Server

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	React JS for frontend form rendering; Flask or Express.js for backend API	React JS, Flask / Express.js
2	Security Implementations	Student login with JWT tokens; input validation to prevent SQL injection; HTTPS enforced	JWT, HTTPS, Input Sanitization
3	Scalable Architecture	3-tier architecture (Presentation, Business Logic, Data) allows independent scaling of each layer	3-tier / Microservices
4	Availability	Load balancer distributes traffic; database replication ensures no single point of failure	Nginx Load Balancer, DB Replication
5	Performance	Caching of book catalogue data; asynchronous notification dispatch to avoid blocking request submission	Redis Cache, Async Task Queue (Celery)

References:

https://c4model.com/
https://developer.ibm.com/patterns/
https://www.ibm.com/cloud/architecture
https://aws.amazon.com/architecture